Online Shopping System — ERP Module Report

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Module implemented: Online Shopping System" Sales & Inventory

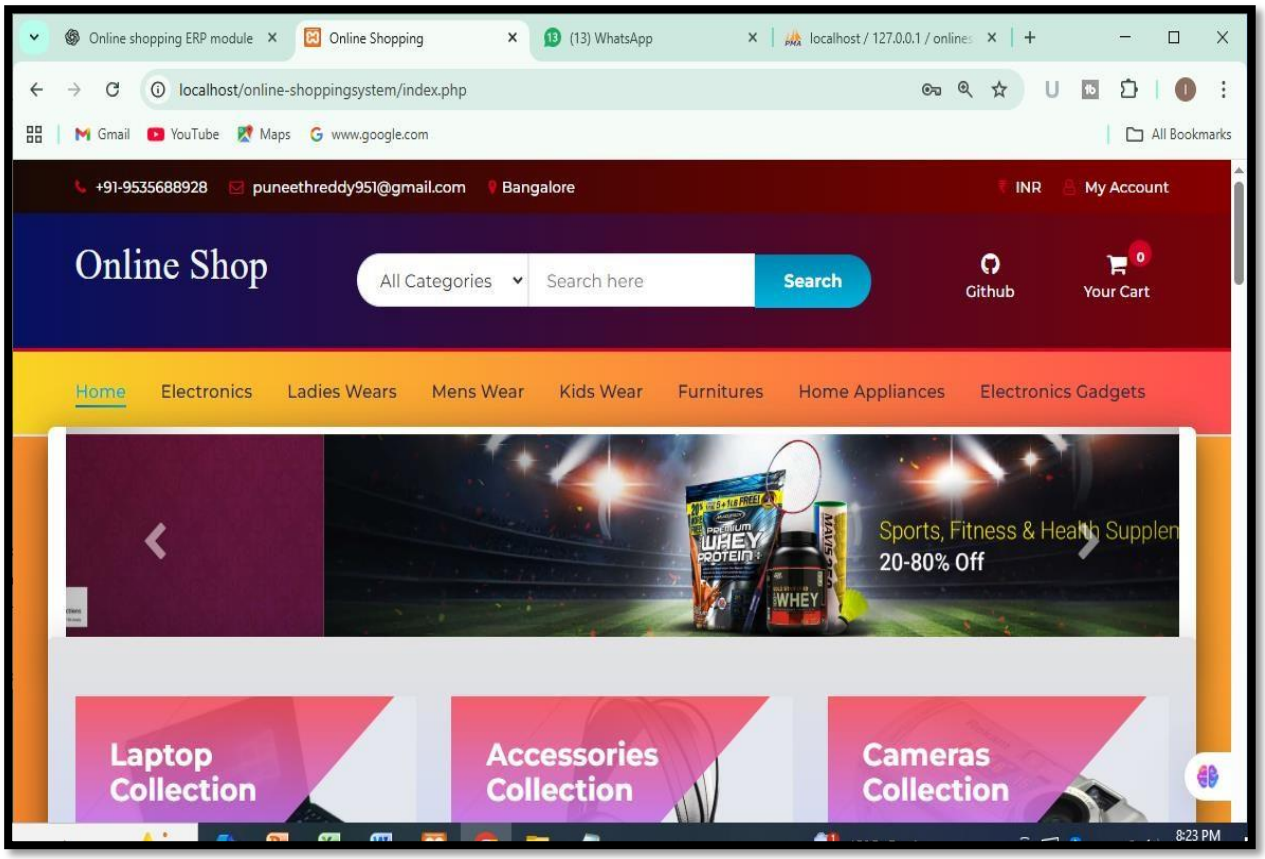
1. Business Problem / Scenario Explanation

Small-to-medium online retailers often struggle to keep sales, stock, and order processing synchronized. Typical pain points:

- Inventory quantities go out of sync when orders are placed.
- Slow manual order processing and invoicing.
- Difficulty tracking order status and customer details.
- No simple report to show low-stock items or daily sales.

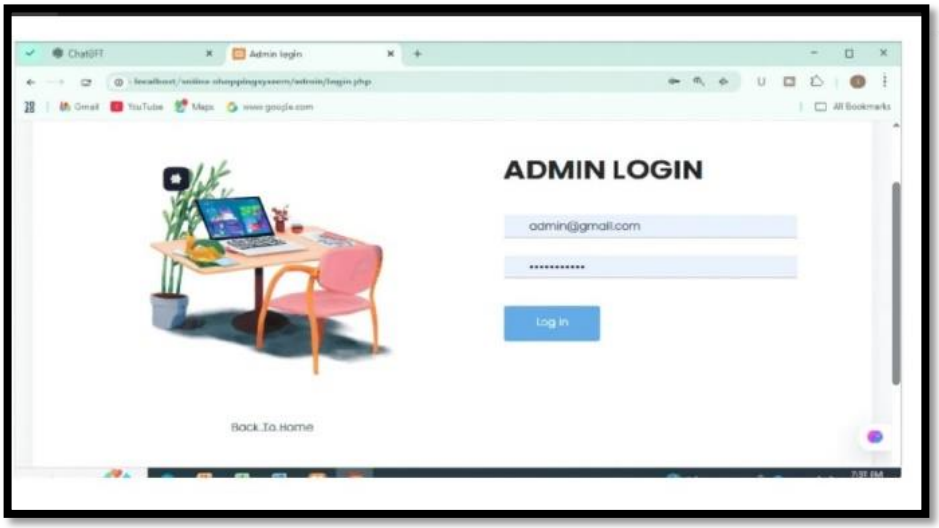
Goal: Build an ERP module (Sales & Inventory) to automate order placement, update inventory, manage customers and products, and provide basic reports. This module will integrate into an

Online Shopping System.



Core Features:

1. **Customer Management**
    a. Add / Edit / Delete customers
    b. Fields: CustomerID, FirstName, LastName, Email, Phone, Address, City, PostalCode, Country, CreatedAt
2. **Order Management (Sales)**
    a. Create new orders (cart -> checkout)
    b. Fields: OrderID, CustomerID, OrderDate, Status (Pending, Processing, Shipped,
        i. Delivered, Cancelled), TotalAmount, PaymentMethod, ShippingAddress
    c. OrderItems: OrderItemID, OrderID, ProductID, Quantity, UnitPrice, LineTotal
3. **Inventory Update**
    a. On order confirmation, reduce product QuantityInStock
    b. Prevent checkout if QuantityInStock < requested quantity
    c. Low stock alerts (QuantityInStock <= ReorderLevel)
4. **Billing / Invoicing**
    a. Generate simple invoice per order (order summary)
5. **Reports**
    a. Daily / Monthly sales summary
    b. Product stock report
    c. Low-stock report
    d. Top-selling products
6. **Authentication (Basic Login)**
    a. Admin login to access dashboard and module
    b. Fields: UserID, Username, PasswordHash, Role



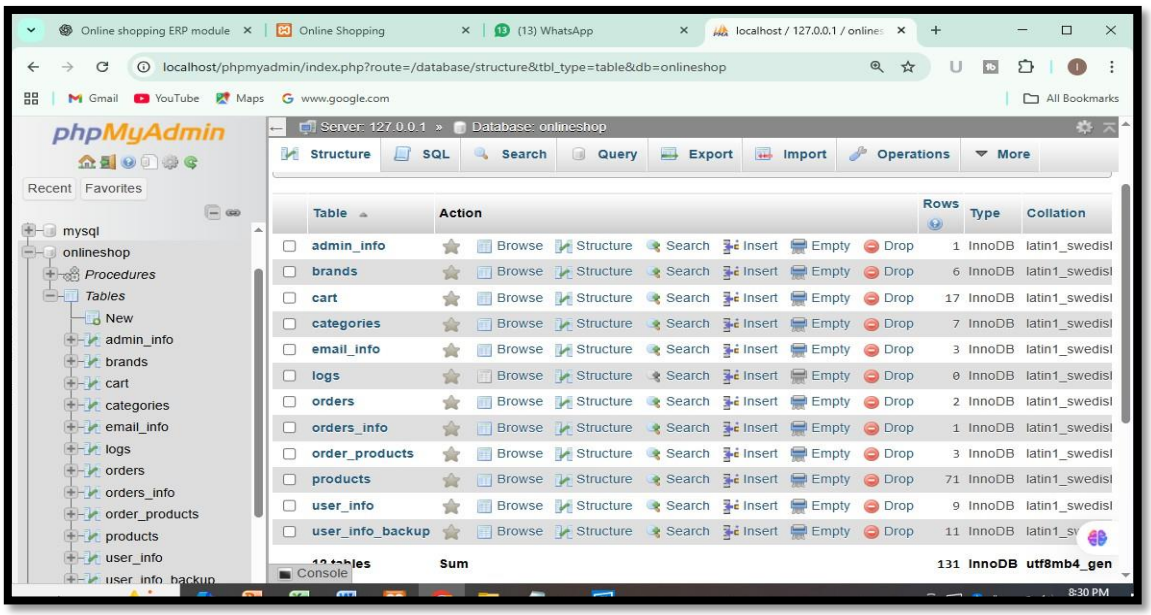
3. Required Data Fields (Detail)

Users(admin)

- user_id (PK)
- username
- password_hash
- role
- created_at

Products

- product_id (PK)
- sku (unique)
- name
- description
- category
- price (decimal)
- cost_price (decimal)
- quantity_in_stock (int)
- reorder_level (int)
- image_path
- status (enum: active/inactive)
- created_at



- first_name
- last_name
- email
- phone
- address
- city
- postal_code
- country
- created_at

Orders

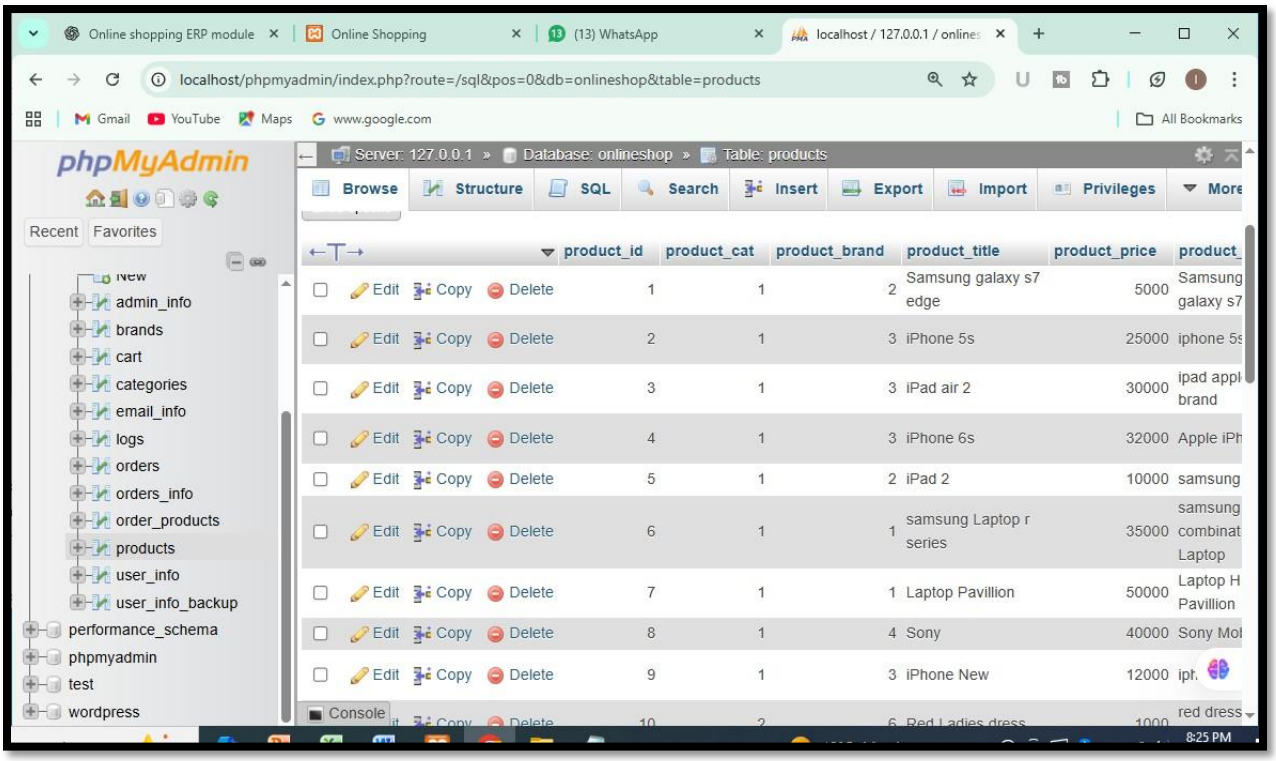
- order_id (PK)
- customer_id (FK -> customers.customer_id)
- order_date (datetime)
- status
- total_amount (decimal)
- payment_method
- shipping_address
- created_at

Order_items

- order_item_id (PK)
- order_id (FK -> orders.order_id)
- product_id (FK -> products.product_id)
- quantity (int)
- unit_price (decimal)
- line_total (decimal)

Inventory_transactions (optional but useful)

- txn_id (PK)
- product_id (FK)
- change_qty (int)
- txn_type (sale, restock, adjustment)
- reference (e.g., order_id)
- created_at



4. Database Design / ERD

Below is a textual ERD (use any diagram tool to draw this as an image):

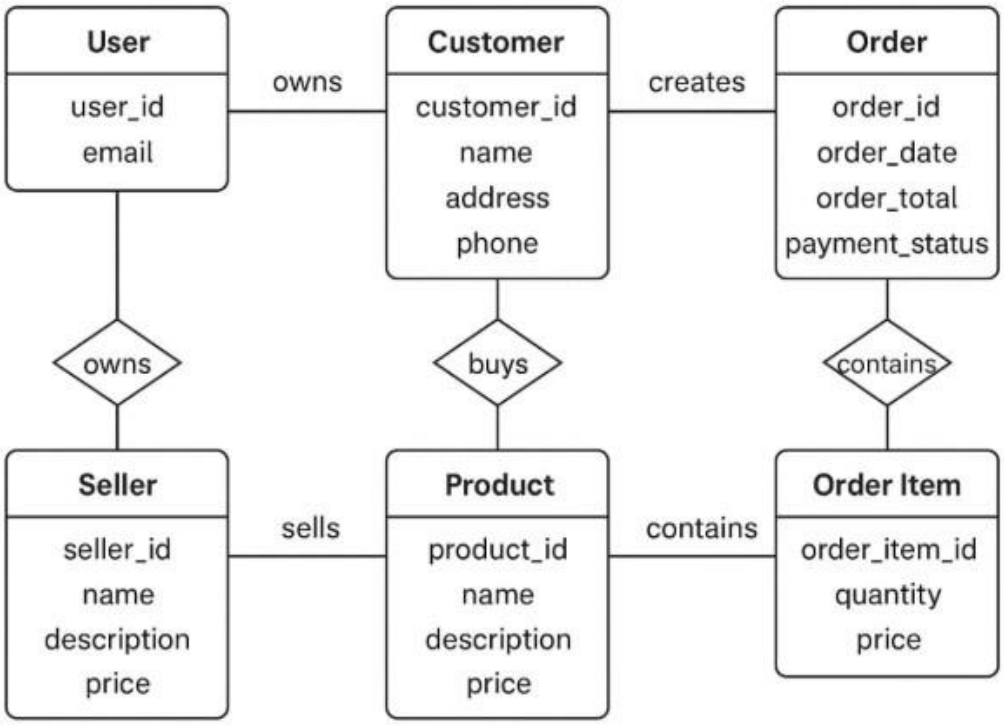
users (1) --- (n) orders
customers (1) --- (n) orders
orders (1) --- (n) order_items
products (1) --- (n) order_items
products (1) --- (n) inventory_transactions

Primary keys:

- users.user_id, customers.customer_id, products.product_id, orders.order_id, order_items.order_item_id, inventory_transactions.txn_id

Foreign keys:

- orders.customer_id -> customers.customer_id
- order_items.order_id -> orders.order_id
- order_items.product_id -> products.product_id
- inventory_transactions.product_id -> products.product_id


User
user_id
email
owns
Customer
customer_id
name
address
phone
creates
Order
order_id
order_date
order_total
payment_status
owns
buys
contains
Seller
seller_id
name
description
price
sells
Product
product_id
name
description
price
contains
Order Item
order_item_id
quantity
price

Technologies Used

1. Programming Languages

- **HTML5** – used for creating the front-end structure of the Online Shopping System.
- **CSS3** – used for designing layouts, styling pages, and making the UI responsive.
- **JavaScript** – used for form validation, dynamic content, and interactive elements.
- **PHP** – used for backend logic, processing requests, and connecting to the database.
- **SQL (MySQL Queries)** – used for data storage, retrieval, updating, and deletion.

2. Frameworks / Tools

- **XAMPP** – for running Apache server and MySQL database locally.
- **phpMyAdmin** – used to create, view, and manage database tables.
- **Bootstrap (optional)** – for responsive and predesigned UI components.
- **Vibe Coding / Code Editor** – any editor such as Sublime Text/VS Code for writing code.

3. Database

- **MySQL** – used as the relational database management system to store all application data (users, products, carts, orders, categories, etc.)

Pros and Cons of ERP Module (Online Shopping System)

✔ Pros

1. **Centralized Data**
   All customer, product, order, and cart information is stored in a single database, making management easy.
2. **Automation**
   The system automatically handles tasks like order placement, cart calculation, stock updates, and user authentication.
3. **User-Friendly Interface**
   Easy navigation, product browsing, and order placement improve the customer experience.
4. **Faster Operations**
   Manual work is reduced—no need for handwritten records or Excel sheets.
5. **Scalable**
   New categories, products, and admin features can be added easily.

6. **Secure Login System**
   Admin and users are authenticated before accessing the system.

✖ Cons

1. **Requires Internet/Local Server**
   Without XAMPP or server hosting, the system will not run.
2. **Technical Knowledge Needed**
   Database management and PHP coding require basic IT skills.
3. **Limited Security (If using MD5 hashing)**
   Password security may be weak unless upgraded to stronger hashing methods.
4. **Single-Admin Limitation**
   The system may require upgrading if multiple roles (admin, manager, staff) are needed.
5. **No Mobile App**
   It works only in a browser; mobile app support must be developed separately.

Conclusion: How ERP System Solves the Business Problem

The **Online Shopping ERP Module** successfully solves the major challenges faced in manual shopping and store management. It automates core processes such as product listing, order placement, cart management, payment handling (cash on delivery), and customer registration. All business data is stored securely in a MySQL database, making it easy to track orders, manage inventory, and analyze sales.

The system reduces human error, improves customer experience, and speeds up business transactions. By integrating frontend pages with backend logic and a relational database, this ERP module provides a complete digital solution for small and medium online stores. It also offers scalability, meaning more features like reports, analytics, and payment gateways can be added in the future.

Overall, the system transforms the traditional shopping workflow into a fast, automated, and user-friendly digital process that benefits both customers and the business owner.